

Abstract

Chronic neuropathic pain is a persistent neurological condition that substantially degrades quality of life, yet its clinical assessment remains largely dependent on subjective self-report scales. The subjective nature of pain, inter-individual variability, and the absence of reliable objective biomarkers make robust pain quantification difficult in both research and clinical practice.

Electroencephalography (EEG) provides a non-invasive and temporally precise measure of brain activity and has shown promise for characterizing pain-related neurophysiological signatures. However, a large portion of prior EEG-based pain literature is constrained by two major issues: (i) the use of computationally expensive nonlinear features (e.g., approximate entropy and sample entropy), which limits feasibility for real-time applications, and (ii) insufficient validation rigor, where non-subject-independent data splitting can inflate reported performance and reduce clinical reliability.

In this work, we present a **computationally efficient, subject-generalizable machine learning framework** for **three-class chronic neuropathic pain classification** (low, moderate, high) using resting-state EEG. The proposed pipeline combines robust EEG preprocessing—including eyes-open (EO) and eyes-closed (EC) separation, artifact suppression via independent component analysis (ICA), and band-limited filtering—with a carefully designed feature engineering strategy that emphasizes **low-complexity yet discriminative descriptors**. Specifically, we extract a broad set of features spanning statistical, time-domain, frequency-domain, time–frequency (wavelet energy), and nonlinear dynamics, including Hjorth parameters, spectral entropy, permutation entropy, differential entropy, and Higuchi fractal dimension. Importantly, these features are chosen to provide strong representational power while maintaining practical computational cost, supporting scenarios where near real-time inference is required.

We train and evaluate a diverse set of classical ML models, including linear methods, tree-based ensembles, boosting approaches, kernel machines, and shallow neural networks. All experiments follow a strict **subject-level train/test protocol** to ensure that the evaluated performance reflects generalization to unseen individuals. In addition, we perform **channel-wise model training** to quantify electrode-specific discriminability and to identify the most informative EEG channels for pain stratification. Our results reveal a consistent subset of channels that yields the strongest performance, with top-performing channels including **M2, P3, O2, P8, and P4**, achieving accuracies of **(xx, xx, xx, xx, xx%)**, respectively, under the best matching model configurations. These findings demonstrate that accurate and deployable chronic pain classification can be achieved using a purely ML-based approach grounded in efficient feature design and rigorous validation, while also providing physiologically interpretable insight into channel-level contributions.

1. Introduction

Chronic neuropathic pain is a complex and persistent condition caused by injury or dysfunction in the somatosensory nervous system. Unlike acute pain, which typically reflects tissue damage and resolves with healing, chronic neuropathic pain can continue for months or years and often lacks a direct measurable physiological correlate. This makes it one of the most challenging conditions to quantify, monitor, and treat effectively. In current clinical practice, pain intensity and progression are primarily assessed using self-report scales such as the numerical rating scale (NRS) or visual analog scale (VAS). Although these tools are widely used due to their simplicity, they are inherently subjective and are strongly influenced by attention, stress, emotional state, cultural background, and

individual tolerance. Consequently, the same objective neural state can yield different reported scores across subjects, and conversely, similar scores can correspond to different neurophysiological states. This mismatch has motivated research toward objective biomarkers that can complement subjective scales and improve reliability in diagnosis, treatment personalization, and outcome tracking.

EEG has emerged as a compelling candidate for objective pain assessment due to its high temporal resolution, portability, and non-invasive nature. A growing body of research suggests that pain perception and chronic pain states modulate oscillatory activity across multiple frequency bands and alter functional dynamics in distributed cortical networks. These alterations can manifest as changes in spectral power, band ratios, temporal variability, and nonlinear complexity—features that are measurable from EEG even during resting-state recordings. Resting-state EEG is especially attractive in clinical environments because it requires minimal task design, reduces subject burden, and can be acquired relatively quickly. Nevertheless, despite the promise of EEG for pain quantification, research in chronic neuropathic pain remains limited and methodologically inconsistent compared to mature EEG applications such as epilepsy analysis or brain–computer interfaces.

A key barrier to translational progress is that many existing EEG-based pain studies rely on nonlinear features that are computationally expensive and sensitive to parameter choices. In particular, approximate entropy (ApEn) and sample entropy (SampEn) are frequently employed to describe irregularity or complexity of EEG time series. While these measures can be discriminative, their computational cost increases substantially with signal length and parameter configurations, often making them impractical for real-time use. Additionally, entropy measures can become unstable under noise, short windows, or inter-subject differences unless carefully tuned. From a deployment perspective, a clinically useful pain monitoring system should operate efficiently, generalize reliably across patients, and remain robust under realistic recording conditions. These requirements motivate the exploration of alternatives that maintain discriminability without imposing high computational overhead.

A second major limitation concerns evaluation methodology—specifically, the absence of strict subject-level validation in many studies. EEG signals contain strong subject-specific signatures linked to skull geometry, electrode placement variation, baseline neural rhythms, and idiosyncratic noise patterns. If window-level or segment-level random splitting is used, data from the same subject can appear in both training and testing sets, enabling the model to exploit subject identity rather than true pain-related patterns. This issue can drastically inflate accuracy and create an illusion of generalization. In contrast, **subject-independent evaluation**—where all samples from a test subject are entirely held out during training—is the appropriate choice for clinically meaningful performance estimates, as it reflects the real-world scenario of predicting pain levels for previously unseen individuals.

In this paper, we address these limitations by presenting a purely machine learning–based EEG framework for **three-class chronic neuropathic pain classification (low, moderate, high)** that is explicitly designed for computational efficiency and subject-generalizable evaluation. Our pipeline begins with standardized EEG preprocessing that includes separation of eyes-open (EO) and eyes-closed (EC) conditions, artifact suppression using independent component analysis (ICA), and band-limited filtering to remove slow drifts and high-frequency noise. We then perform window-based feature extraction using a broad but efficient set of descriptors spanning multiple domains: (i) statistical and distributional measures (e.g., trimmed means, MAD, percentiles), (ii) time-domain dynamics (e.g., zero-crossing rate, Teager–Kaiser energy, autocorrelation), (iii) spectral bandpower features computed via Welch’s method across delta–gamma bands and band ratios, (iv) Hjorth

activity–mobility–complexity parameters, (v) spectral shape descriptors such as spectral centroid, spectral entropy, and spectral edge frequency, (vi) wavelet energy features for multi-resolution time–frequency characterization, and (vii) nonlinear features selected for favorable computational profiles, including permutation entropy, differential entropy, and Higuchi fractal dimension. This design retains the ability to capture both linear and nonlinear EEG dynamics while remaining more practical for scalable and real-time systems than approaches centered on ApEn/SampEn.

To ensure robustness and broad applicability, we evaluate a diverse family of classical ML models, including linear classifiers, tree-based ensembles, boosting frameworks, kernel-based methods, and shallow neural networks. Most importantly, we enforce a strict subject-level train/test split throughout all experiments, providing an unbiased estimate of generalization to unseen individuals. Beyond aggregated modeling across all channels, we also conduct **channel-wise training**, where each EEG channel is trained independently using the full feature set, enabling electrode-level discriminability analysis. This channel-wise view is particularly valuable for identifying compact electrode subsets that may be used in practical acquisition systems, where minimizing electrode count can reduce setup time, increase comfort, and improve clinical feasibility.

Our experimental findings highlight that a small subset of electrodes consistently provides strong discrimination performance for neuropathic pain stratification. In particular, the channels **M2, P3, O2, P8, and P4** emerge as top-performing channels under the best matching model configurations, achieving accuracies of (**xx, xx, xx, xx, xx%**), respectively. These results suggest that meaningful pain-related information is captured in specific scalp regions and that efficient ML-based pipelines can achieve competitive performance without heavy deep learning or high-complexity entropy features.

Contributions. The primary contributions of this paper are:

1. A **pure ML EEG pain classification framework** emphasizing **low-complexity nonlinear and spectral features** suitable for real-time use.
2. A rigorously designed evaluation using **subject-level validation** for realistic generalization.
3. A systematic **channel-wise analysis** to identify a compact subset of informative electrodes for chronic neuropathic pain stratification.
4. A comparative study across multiple ML model families to determine robust classifiers for this task.

6. Conclusion

This work presents a computationally efficient, subject-generalizable machine learning pipeline for three-class chronic neuropathic pain classification using EEG. By prioritizing low-complexity feature design and rigorous validation, we address two common limitations in prior EEG-based pain studies: reliance on high time-complexity nonlinear measures and evaluation protocols that do not reflect true subject-independent generalization. Our approach integrates standardized EEG preprocessing—including EO/EC separation, ICA-based artifact suppression, and band-limited filtering—with a multi-domain feature set spanning statistical, temporal, spectral, time–frequency, and nonlinear dynamics while avoiding computationally heavy entropy measures such as ApEn and SampEn.

A core emphasis of this study is realistic deployment potential. The extracted features—such as Hjorth parameters, permutation and differential entropy, spectral entropy, wavelet energy, and fractal descriptors—capture discriminative pain-related structure while remaining practical for

scalable inference. Extensive experimentation across diverse classical ML families demonstrates that reliable pain stratification can be achieved without deep learning, provided that feature engineering is thoughtfully constructed and validated under strict subject-level splits.

Beyond overall classification performance, the channel-wise training strategy provides physiologically meaningful insight and supports practical EEG system design. Our analysis identifies a consistent subset of electrodes—**M2, P3, O2, P8, and P4**—as particularly informative for distinguishing low, moderate, and high pain states, achieving accuracies of (**xx, xx, xx, xx, xx%**) under optimal model selection. This suggests that accurate pain monitoring may be feasible even with reduced channel configurations, improving usability and comfort in clinical or ambulatory settings.

Overall, the proposed ML-based framework contributes a robust foundation for objective EEG-based chronic pain assessment, balancing computational efficiency, interpretability, and realistic subject-independent performance. Future extensions can explore cross-dataset generalization, domain adaptation to reduce subject variability, longitudinal modeling for tracking pain progression, and clinically oriented validation protocols to further bridge the gap between research and real-world deployment.

Methodology

1. Feature Creation

1.1 Signal segmentation and labeling

EEG recordings were processed at a sampling frequency of **250 Hz** and segmented into fixed-length windows of **5 seconds** (i.e., **1250 samples per window**) with **50% overlap** to increase sample density while maintaining temporal continuity.

Each EEG recording was associated with a subject-specific pain score, which was mapped into a **three-class pain stratification** problem: **low**, **mid**, and **high** pain. In the final setting used for this work, windows from neutral/ambiguous pain (depending on the labeling version) were excluded to reduce label noise and to maintain consistent class semantics for supervised learning.

This design aligns with the practical goal of pain stratification in clinical decision making, where coarse-grained pain levels are often more actionable than fine-grained regression targets.

1.2 Multi-domain feature set design principles

A key objective of this work is to construct a feature set that is both **discriminative** and **computationally efficient**, enabling future deployment in real-time or near-real-time settings.

Therefore, we deliberately avoid high time-complexity nonlinear measures such as **Approximate Entropy (ApEn)** and **Sample Entropy (SampEn)**, which often require expensive pattern comparisons and become costly as window length increases. Instead, we focus on features that can be computed using efficient operations (e.g., Welch PSD estimation, simple statistics, wavelet decomposition, short-order permutation counts, and low-k fractal approximations) while still capturing pain-relevant neurophysiological information.

The extracted features span complementary domains to capture:

- (1) distributional changes (amplitude variability),
- (2) temporal dynamics (signal transitions and local energy),
- (3) spectral structure (band-limited oscillatory modulation),
- (4) time–frequency behavior (multi-resolution energy allocation), and
- (5) nonlinear complexity (irregularity and fractal-like properties).

1.3 Statistical features

We compute standard descriptive statistics such as mean, variance, standard deviation, min/max, median, skewness, and kurtosis to capture changes in EEG amplitude distribution that may reflect altered cortical excitability under different pain intensities.

To improve robustness to non-Gaussian artifacts and heavy-tailed distributions (common in EEG), we additionally compute trimmed means (10% and 15%), median absolute deviation (MAD), and interquartile range (IQR). Percentile-based descriptors (P10, P25, P75) provide stable distribution summaries that are less sensitive to transient spikes.

Why these help for pain: chronic pain has been associated with altered resting-state activity and variability; distributional descriptors provide low-cost signals of these shifts and often serve as strong baselines in EEG classification.

1.4 Time-domain dynamic features

Time-domain features quantify the local temporal behavior of EEG windows. We compute first-difference statistics (mean and standard deviation of the derivative), mean absolute difference, maximum absolute difference, and sign-change rate to capture rapid fluctuations and transition behavior.

We also compute zero-crossing count as a proxy for oscillatory behavior and waveform complexity, lag-1 autocorrelation to capture short-range temporal dependence, and the **Teager–Kaiser energy operator**, which is a lightweight nonlinear energy estimate sensitive to instantaneous amplitude–frequency interactions.

Why these help for pain: pain-related cortical states can alter EEG rhythm regularity and transient activity; time-domain features provide a computationally cheap way to detect such differences without requiring deep models.

1.5 Frequency-domain bandpower and ratios

We compute bandpower using Welch’s PSD estimate across canonical EEG bands: **delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–45 Hz)**. Both absolute and relative bandpowers are extracted, along with clinically interpretable ratios (theta/alpha, theta/beta, alpha/beta).

Why these help for pain: multiple studies have reported pain-associated modulations in alpha and beta rhythms as well as shifts in slower activity; ratios provide normalization against overall power variations and can reflect systematic changes in oscillatory balance.

1.6 Hjorth parameters

Hjorth activity, mobility, and complexity are computed to summarize signal variance, dominant frequency tendency, and waveform complexity in a compact manner. These features are widely used in EEG because they are informative yet computationally inexpensive, requiring only low-order differences and variance estimates.

Why these help for pain: they capture both amplitude and morphological characteristics that can shift with cortical state changes, including those linked to pain perception.

1.7 Spectral shape descriptors

Beyond bandpower, we compute spectral centroid (power-weighted mean frequency), spectral entropy (normalized PSD entropy), and the 90% spectral edge frequency (SEF90), which collectively characterize the distribution and concentration of spectral power.

Why these help for pain: pain can alter how power is distributed across frequencies (e.g., more concentrated vs. more spread); these descriptors provide a global spectral summary.

1.8 Wavelet energy features (time–frequency)

To capture non-stationary time–frequency behavior, we perform discrete wavelet decomposition (db4, level=4) and compute energy at each decomposition level. Wavelet energies provide multi-resolution characterization while remaining computationally practical for short windows.

Why these help for pain: EEG under pain may show transient dynamics and scale-specific energy shifts that pure bandpower can miss.

1.9 Efficient nonlinear complexity features

We include nonlinear features selected for favorable computational profiles:

- **Permutation entropy** (order=3, delay=1) captures ordinal-pattern irregularity using low-order permutations, making it substantially cheaper than ApEn/SampEn while still reflecting complexity changes.
- **Differential entropy** (Gaussian approximation) provides a closed-form complexity measure based on variance, offering a lightweight proxy for information content.
- **Higuchi fractal dimension (HFD)** estimates fractal-like complexity using small k ($k_{\max}=10$), enabling efficient computation on short windows.

Why these help for pain: chronic pain is often linked to altered cortical complexity and irregularity; these efficient nonlinear measures provide a practical compromise between discriminative power and runtime cost.

2. Feature Selection

2.1 Motivation

The extracted feature set is intentionally broad and multi-domain, which increases representational power but also introduces redundancy and potential overfitting—especially in EEG where inter-subject variability is high. Feature selection serves three goals in this work:

- (1) improving generalization by reducing spurious correlations,
- (2) enhancing interpretability by identifying stable pain-relevant descriptors, and
- (3) supporting deployment by minimizing computational load at inference time.

2.2 Hybrid feature selection strategy

We adopt a **hybrid ranking approach** that aggregates feature importance scores from complementary selection families: **filter**, **embedded**, and **wrapper** methods. The central rationale is

that EEG features can exhibit both linear and nonlinear relationships with pain labels, and no single selection method is uniformly reliable across all signal conditions. Therefore, we compute feature scores using multiple methods and then form a consensus importance via normalized score averaging.

2.3 Filter methods

Two filter-based relevance measures are applied:

- **ANOVA F-test (f_classif)** ranks features based on between-class mean separation relative to within-class variance, favoring features with strong class discrimination under approximately linear separability assumptions.
- **Mutual information (mutual_info_classif)** captures nonlinear dependency between individual features and labels, making it suitable when relationships are non-monotonic or non-linear.

Why filters help here: EEG datasets often have many correlated features; filter methods are fast, model-agnostic, and provide a stable first-pass ranking without overfitting to a particular classifier.

2.4 Embedded methods

We compute importance scores from models that learn feature relevance as part of training:

- **Random Forest feature importance** captures nonlinear interactions and ranks features by impurity reduction in tree ensembles.
- **XGBoost feature importance** provides gradient-boosted tree-based relevance, often more sensitive to subtle predictive structure than bagging-based forests.
- **Logistic regression coefficient magnitude** (after standard scaling) captures stable linear contributions and often yields interpretable markers.

Why embedded helps here: pain-related EEG patterns may be nonlinear and interaction-dependent, where tree ensembles can uncover relationships that simple filters miss, while logistic regression can detect robust linear effects that generalize well.

2.5 Wrapper method

We apply **Recursive Feature Elimination (RFE)** with logistic regression as a wrapper selection method. RFE iteratively removes features that contribute least to predictive performance under a trained model, producing a ranking that is more performance-driven than filter methods.

Why wrapper helps here: wrapper methods directly optimize for predictive utility, often improving performance when feature redundancy exists, albeit at higher computational cost (still acceptable in offline training).

2.6 Score normalization and aggregation

Because each selection method outputs scores on different scales, we normalize each score vector to [0,1] and compute an **average consensus score** across methods. Features are then ranked by this average score, and the top-ranked subset is retained (in your current implementation, the top features per channel are summarized and saved). This consensus strategy reduces the risk of selecting unstable features that appear important only under one criterion.

3. ML-based Pipeline

3.1 Channel-wise dataset construction

To study electrode-specific discriminability and to enable reduced-channel system design, we follow a per-channel pipeline. For each EEG channel, subject-wise windowed feature CSVs are merged into a channel-level dataset, producing one consolidated feature table per channel.

Non-informative constant features are removed to prevent degenerate behavior and to reduce noise in model fitting.

3.2 Subject-level train/test split

A strict **subject-stratified split** is applied, ensuring that all windows from a given subject appear exclusively in either train or test sets. This prevents information leakage and yields a realistic evaluation of generalization to unseen individuals.

Stratification is performed at the subject level to maintain balanced label distributions across splits as far as possible, which is critical for stable macro-F1 estimates in three-class settings.

3.3 Feature scaling

For models sensitive to feature scaling (e.g., logistic regression, SVM, kNN, MLP, SGD), we apply **standardization** based on the training set statistics and reuse the same transformation on the test set to avoid leakage.

3.4 Model zoo and evaluation protocol

We evaluate a diverse set of ML classifiers representing different inductive biases:

- **Linear models:** Logistic Regression, SGDClassifier
- **Tree-based models:** Decision Tree, Random Forest
- **Boosting ensembles:** Gradient Boosting, AdaBoost, XGBoost, LightGBM
- **Other models:** RBF-SVM, kNN, Gaussian Naive Bayes, MLPClassifier

This diversity is intentional: pain-related EEG structure may be partially linear (captured by LR/SGD), partially nonlinear with interactions (captured by forests/boosting), or margin-separable in high-dimensional space (captured by RBF-SVM). Using multiple model families also reduces dependence on a single classifier assumption and strengthens the evidence that improvements arise from the feature design rather than model overfitting.

3.5 Metrics and best-model selection per channel

For each channel, all models are trained on the subject-independent training set and evaluated on the held-out subject test set. Performance is summarized using **accuracy** and **macro-F1**, and detailed class-wise precision/recall/F1 are saved via a classification report along with a confusion matrix visualization.

The best-performing model per channel is selected based on maximum accuracy (as implemented), and an overall summary file is generated listing each channel's best model and accuracy, enabling ranking of channels by discriminability.

3.6 Rationale for channel-wise learning

Channel-wise training serves both scientific and engineering goals. Scientifically, it helps identify scalp regions whose activity most consistently differentiates pain levels, supporting interpretability. Practically, it enables selection of a compact electrode subset for future deployment, reducing acquisition overhead and improving usability, which is essential for any clinical pain monitoring system.
